

Electronics Assignment

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Submitted on: 27 April 2026

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1. Multiplexer (4-to-1 MUX)

Problem Statement

Construct a 4-to-1 Multiplexer capable of choosing one among four data input lines (I0, I1, I2, I3) and forwarding it to a single output Y, controlled by two selection lines S1 and S0.

Inputs and Outputs

Inputs : I0, I1, I2, I3 — Data lines | S1, S0 — Selection lines

Output : Y — Selected output

Truth Table

S1	S0	Output Y
0	0	Y = I0
0	1	Y = I1
1	0	Y = I2
1	1	Y = I3

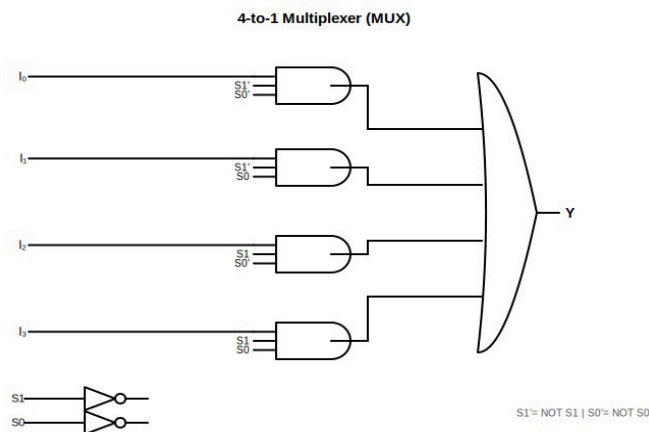
Boolean Expression

$$Y = (S1' \cdot S0' \cdot I0) + (S1' \cdot S0 \cdot I1) + (S1 \cdot S0' \cdot I2) + (S1 \cdot S0 \cdot I3)$$

Reduction

Minimization is not applicable here. Every minterm is an essential prime implicant, each covering a distinct selection combination.

Circuit Diagram



2. De-Multiplexer (1-to-4 DEMUX)

Problem Statement

Design a 1-to-4 De-Multiplexer that routes a single input D to one of four output lines Y0–Y3 based on two selection lines S1 and S0.

Inputs and Outputs

Inputs : D — Data input | S1, S0 — Selection lines

Outputs : Y0, Y1, Y2, Y3 — Output lines (only one active at a time)

Truth Table

S1	S0	Y0	Y1	Y2	Y3
0	0	D	0	0	0
0	1	0	D	0	0
1	0	0	0	D	0
1	1	0	0	0	D

Boolean Expression

$$Y0 = D \cdot S1' \cdot S0'$$

$$Y1 = D \cdot S1' \cdot S0$$

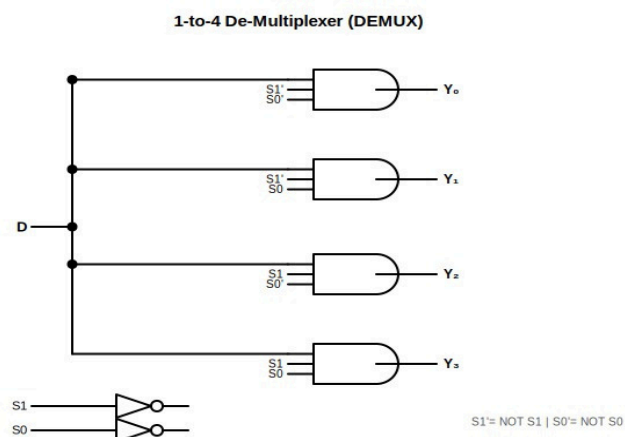
$$Y2 = D \cdot S1 \cdot S0'$$

$$Y3 = D \cdot S1 \cdot S0$$

Reduction

Minimization is not applicable here. Each output maps to a unique minterm that requires all three input variables.

Circuit Diagram



3. Encoder (4-to-2 Line Encoder)

Problem Statement

Design a 4-to-2 Encoder that accepts four input lines I_0 – I_3 (only one HIGH at a time) and produces a 2-bit binary output code A_1, A_0 corresponding to the active input.

Inputs and Outputs

Inputs : I_0, I_1, I_2, I_3 — Input lines (only one active HIGH at a time)

Outputs : A_1 (MSB), A_0 (LSB) — 2-bit encoded output

Truth Table

I_3	I_2	I_1	I_0	A_1	A_0
0	0	0	1	0	0
0	0	1	0	0	1
0	1	0	0	1	0
1	0	0	0	1	1

Boolean Expression

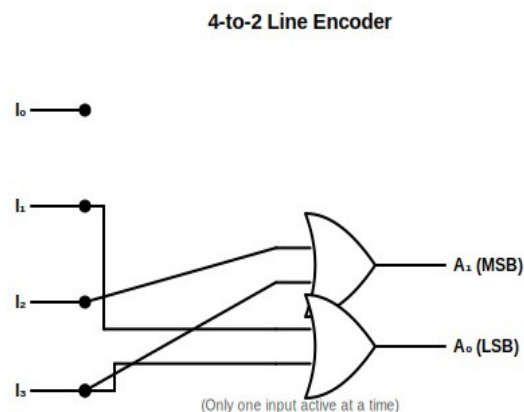
$$A_1 = I_2 + I_3$$

$$A_0 = I_1 + I_3$$

Reduction

Both expressions are already in their simplest form. A_1 encompasses positions 2 and 3 where the MSB equals 1; A_0 covers positions 1 and 3 where the LSB equals 1. No additional K-Map simplification is achievable.

Circuit Diagram



4. Decoder (2-to-4 Line Decoder)

Problem Statement

Design a 2-to-4Decoder that accepts a 2-bit binary input A1, A0 and activates exactly one of four output lines Y0–Y3 correspondingto the decimal value of the input.

Inputs and Outputs

Inputs : A1 (MSB),A0 (LSB) — 2-bit binary input

Outputs : Y0 (decimal0), Y1 (decimal 1), Y2 (decimal 2), Y3 (decimal 3)

Truth Table

A1	A0	Y0	Y1	Y2	Y3
0	0	1	0	0	0
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	1

Boolean Expression

$$Y0 = A1' . A0'$$

$$Y1 = A1' . A0$$

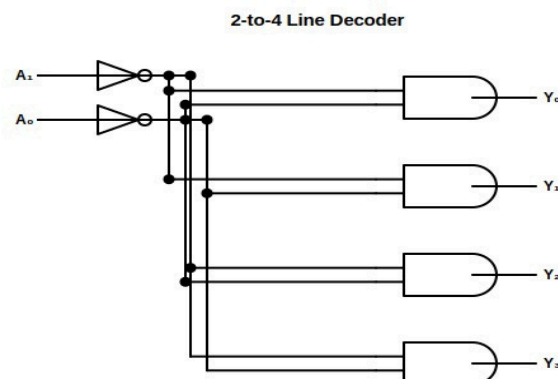
$$Y2 = A1 . A0'$$

$$Y3 = A1 . A0$$

Reduction

Noreductionpossible. Each output is a single minterm. All literals are essential — removing any causes multiple outputs to activate.

Circuit Diagram



5. Parity Generator (4-Bit Even Parity)

Problem Statement

Construct a 4-bit EvenParity Generator that takes four data bits D3, D2, D1, D0 and produces a parity bit P ensuring the total number of 1s across the resulting 5-bit word remains even at all times.

Inputs and Outputs

Inputs : D3, D2,D1,D0 — Four data bits

Output : P — Evenparity bit

Truth Table

D3	D2	D1	D0	Count of 1s	P (Even Parity)
0	0	0	0	0 (even)	0
0	0	0	1	1 (odd)	1
0	0	1	0	1 (odd)	1
0	0	1	1	2 (even)	0
0	1	0	0	1 (odd)	1
0	1	0	1	2 (even)	0
0	1	1	0	2 (even)	0
0	1	1	1	3 (odd)	1
1	0	0	0	1 (odd)	1
1	0	0	1	2 (even)	0
1	0	1	0	2 (even)	0
1	0	1	1	3 (odd)	1
1	1	0	0	2 (even)	0
1	1	0	1	3 (odd)	1
1	1	1	0	3 (odd)	1
1	1	1	1	4 (even)	0

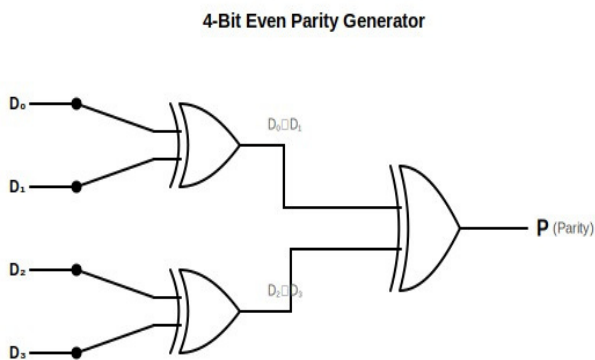
Boolean Expression

$$P = D3XORD2XORD1 XOR D0$$

Reduction

XORparitycreates a checkerboard pattern on the K-Map — no groupings of adjacent 1s exist. The XOR expressionisalready the minimal form.

Circuit Diagram



6. Comparator (2-Bit Magnitude Comparator)

Problem Statement

Construct a 2-bit Magnitude Comparator that evaluates two 2-bit numbers $A = A_1A_0$ and $B = B_1B_0$ and generates three distinct outputs to indicate whether $A > B$, $A = B$, or $A < B$.

Inputs and Outputs

Inputs : A_1, A_0 — First 2-bit number | B_1, B_0 — Second 2-bit number

Outputs : $G (A > B), E (A = B), L (A < B)$ — Comparison result flags

Truth Table

A_1	A_0	B_1	B_0	$G (A > B)$	$E (A = B)$	$L (A < B)$
0	0	0	0	0	1	0
0	0	0	1	0	0	1
0	0	1	0	0	0	1
0	0	1	1	0	0	1
0	1	0	0	1	0	0
0	1	0	1	0	1	0
0	1	1	0	0	0	1
0	1	1	1	0	0	1
1	0	0	0	1	0	0
1	0	0	1	1	0	0
1	0	1	0	0	1	0
1	0	1	1	0	0	1
1	1	0	0	1	0	0
1	1	0	1	1	0	0
1	1	1	0	1	0	0
1	1	1	1	0	1	0

Boolean Expression

Let $x_1 = A_1 \text{XNOR } B_1$ $x_0 = A_0 \text{XNOR } B_0$

$$E (A = B) = x_1 \cdot x_0$$

$$G (A > B) = A_1.B_1' + x_1.A_0.B_0'$$

$$L (A < B) = A_1'.B_1 + x_1.A_0'.B_0$$

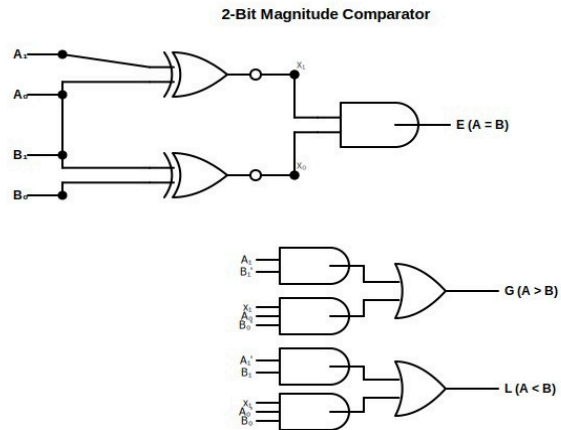
Reduction

$E = x_1 \cdot x_0$ — Already minimal.

$G = A_1.B_1' + x_1.A_0.B_0'$ — Two essential prime implicants, no further reduction.

$L = A_1'.B_1 + x_1.A_0'.B_0$ — Symmetric to G , already minimal.

Circuit Diagram



7. Synchronous Counter (3-Bit Up Counter — JK Flip-Flops)

Problem Statement

Construct a 3-bit synchronous binary up counter built with JK flip-flops that sequences from 000 through 111 and wraps back to 000. Every flip-flop responds simultaneously to the same rising clock edge.

Inputs and Outputs

Input : CLK — Common clock (rising edge triggered)

Outputs : Q2 (MSB), Q1, Q0 (LSB) — 3-bit binary count

Truth Table (State Sequence)

Count	Q2	Q1	Q0	Next Q2	Next Q1	Next Q0
0	0	0	0	0	0	1
1	0	0	1	0	1	0
2	0	1	0	0	1	1
3	0	1	1	1	0	0
4	1	0	0	1	0	1
5	1	0	1	1	1	0
6	1	1	0	1	1	1
7	1	1	1	0	0	0

JK Flip-Flop Excitation Table

Q(t) to Q(t+1)	J	K
0 to 0 (Hold)	0	X
0 to 1 (Set)	1	X
1 to 0 (Reset)	X	1
1 to 1 (Hold)	X	0

X = Don't Care

JK Input Derivation Table

Q2 Q1 Q0	J2	K2	J1	K1	J0	K0
0 0 0	0	X	0	X	1	X
0 0 1	0	X	1	X	X	1

Q2 Q1 Q0	J2	K2	J1	K1	J0	K0
0 1 0	0	X	X	0	1	X
0 1 1	1	X	X	1	X	1
1 0 0	X	0	0	X	1	X
1 0 1	X	0	1	X	X	1
1 1 0	X	0	X	0	1	X
1 1 1	X	1	X	1	X	1

Boolean Expression (After K-Map Reduction)

$J0 = K0 = 1$ ($Q0$ toggles every clock)

$J1 = K1 = Q0$ ($Q1$ toggles when $Q0 = 1$)

$J2 = K2 = Q1.Q0$ ($Q2$ toggles when $Q1 = Q0 = 1$)

Reduction

K-Map for $J0, K0$: Every cell contains 1, yielding $J0 = K0 = 1$

K-Map for $J1, K1$: Output is 1 exclusively when $Q0 = 1$, giving $J1 = K1 = Q0$

K-Map for $J2, K2$: Output is 1 only when both $Q1$ and $Q0$ are 1, resulting in $J2 = K2 = Q1.Q0$

Circuit Diagram

